Klara Nitsche

Atlanta, GA • klin1456@colorado.edu • klaranit.github.io • She/Her

EDUCATION

ATLAS Institute, University of Colorado at Boulder College of Engineering — **Boulder, CO**
Masters of Science in Creative Technology and Design — Cumulative GPA: 3.9/4.0 — *May 2026*

Emory University — **Atlanta, GA**
Bachelor of Arts in Sociology, co-major Integrated Visual Arts — Cumulative GPA: 3.93/4.0 — *May 2024*
Honors: Highest Honors

PUBLICATIONS

[Under Review] **Klara Nitsche,** Ellen Do. 2026. "Marshmallows as Design Tools: From MDD to Marshmallow Robots." In *ACM TEI 2026 Work in Progress paper track.*

[Under Review] Krithik Ranjan, Suibi Weng, **Klara Nitsche**, Ellen Do. 2026 "AI 'Puppet' not 'Agent': Creative, Critical, and Embodied Introduction to AI Chatbots for Elementary Students." In *ACM TEI 2026 Work in Progress paper track,*

Klara Nitsche, Ayesha Rawal, Shalimar Alvarado Cruz, Aaron Neyer, and Cambria Klinger. 2026. "BioTelia: Humans Embodying Pollinators in an Interactive Installation." In *EAI ArtsIT 2026 Conference Proceedings Poster & Demo Track*, Springer. TBP

Klara Nitsche, Ellen Do. 2026. "Marshmallow Robot Skins: Edible Human Robot Interaction" *ACM Creativity & Cognition 2026 Student Volunteer Poster Session.*

RESEARCH EXPERIENCE

Human Interaction and Robotics Group (HIRO), Dept. of Computer Science — **Boulder, CO**
Research Collaborator — *August 2025 - Present*
Under Dr. Alessandro Roncone

- Designed and 3D printed multi-material soft robotic skins with proximity sensors to expand embodied intelligence..
- Built experimental TPU fashion accessories for H1-Unitree Humanoid. Intended to protect falling limbs from body collision during sudden failure.
- Edited manuscripts and produced demonstration videos showing robot task completion for submission to ICRA.

ACME Lab, ATLAS Institute — **Boulder, CO**
Research Collaborator — *August 2025 - Present*
Under Dr. Ellen Yi-Luen Do

- Contributed to *AI Puppet*, a project promoting AI literacy in early education by teaching students the limitations of AI. Paper results currently under review.
- Transcribed and analyzed participatory design workshops with elderly participants on the topic of designing their ideal robots. Ongoing analysis and authorship for pending paper submission.

BA Honor's Thesis — **Atlanta, GA**
Independent Researcher — *May 2024*

- Analyzed 300+ archived local punk posters in the Stuart A. Rose Manuscript and Rare Book Archive at Emory University.
- Awarded the Alan Rackoff Prize for best research usage of archival materials.

Sociology Research Apprenticeship Program, Emory Sociology Department — **Atlanta, GA**
Research Assistant — *August 2022 - December 2022*
Under Dr. Tracy L. Scott

- Collected and curated set of visual data for faculty research project studying astronaut family dynamics and emerging social concepts of science, family, labor, and technology in the 1960's.

TEACHING EXPERIENCE

Object; Introduction to Physical Computing, ATLAS Institute — **Boulder, CO**
Course Assistant — *January 2026 - May 2026*
Under Dr. Michael Rivera

- Taught lab session and graded for 12 undergraduate students on topics of soldering, coding, 3D-printing, circuits, and design..

Materials Studies & Practices, ATLAS Institute — **Boulder, CO**
Course Assistant — *August 2025 - December 2025*
Under Zack Weaver

- Taught class demonstrations on working with materials for mixed undergraduate and graduate-level, 40 person course.
- Led seminar on the meaning of cultural objects in design.

TALKS AND PRESENTATIONS

Materiality of Robot Skin: Lightning Talk to CU Engineering Department **Boulder, CO**
Speaker *September 2025*

- Presented ongoing research of experimental designs for multi-material 3D printed robot skins to audience of 40+ people. Specifically introduced a multi-material dual-layer method where cameras detect indentations on the texturized skin.

Handwriting as an Artifact of Materiality: Presentation to ACME and ATLAS Community **Boulder, CO**
Speaker *September 2025*

- Discussed synthesis of past archival research and introduced a theoretical approach to understanding handwritten artifacts (graffiti, letters, posters, etc.) through a materiality and material-culture lens.

Experimental Protective Fashion: Presentation to HIRO Group **Boulder, CO**
Speaker *September 2025*

- Presented novel TPU wrist and cuff designs for an H1-Humanoid to protect falling limbs from damaging its body in sudden failure. Drew parallels between 3D printing fashion and existing avant-garde collections to highlight intersections in structural design.

AWARDS & FUNDING

ATLAS Distinguished Graduate Student Award **Atlanta, GA**
Recipient *May 2026*

- Awarded to ATLAS students who demonstrate excellence in research, community engagement, creativity and leadership.

Alan Rackoff Prize **Atlanta, GA**
Recipient *May 2024*

- Research prize awarded to the best project, paper, or honor's thesis written on the basis of primary source material from Stuart A. Rose Library collections. Selected by a committee of Emory University librarians, faculty, and community partners. This award's criteria emphasized approach to archival research through strategy, presentation, and analysis of results.

Association of Federal Communications Consulting Engineers (AFCCE) Scholarship **Boulder, CO**
Recipient *August 2025*

- Scholarship for engineering students studying telecommunications technologies awarded on merit and academic engagement.

UNIVERSITY WORK EXPERIENCE

ATLAS Social Recruiter and Ambassador **Boulder, CO**
Employee *August 2025 - August 2026*

- Give tours of ATLAS, host workshops inviting external industry professionals to speak with graduate students, host internal community-centered events for graduate students, and provide student advising for graduate students in ATLAS.

B2 Blackbox Theater Social Media Manager **Boulder, CO**
Employee *August 2024 - August 2026*

- Photograph events, design social media posts, promote events through social media and printed advertisements.

ATLAS Event Photographer **Boulder, CO**
Employee *November 2025 - August 2026*

- Document, edit, and share photography of classes, events, projects, and community assemblies.

COMMUNITY ENGAGEMENT

Buff Food Pantry, CU Boulder **Atlanta, GA**
Volunteer *August 2024 – August 2026*
Awarded Most Active Volunteer

- Re-stock, organize, and facilitate supplies to support community members seeking food security.

WMRE Emory Radio, Emory University **Atlanta, GA**
Creative Board Executive Member, Zine Manager, DJ *February 2022 – May 2024*

- Host weekly DIY workshops for student body to teach accessible, sustainable art practices.
- Design and craft physical merchandise such as t-shirts, sweatpants, zines, and posters for WMRE.
- Co-host weekly show to share new music and spotlight local artists.

SKILLS

Technical: Multi-material 3D printing, welding, soldering, physical computing, Python (PyBullet, Matplotlib)
Analysis: Qualitative data collection and analysis, conducting interviews, archival research, thematic coding, SPSS, MAXQDA
Design: Blender, Solid Works, Rhino, Figma, Adobe Creative Suite
Languages: Fluent in English, Polish, German. Intermediate in Italian and French.